

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAMINATION

January 2026

Probability (Ph.D. Version)

Instructions to the Student

- Answer all six questions. Each will be graded from 0 to 10.
- Use a different booklet for each question. Write the problem number and your code number (NOT YOUR NAME) on the outside cover.
- Keep scratch work on separate pages in the same booklet.
- If you use a “well known” theorem in your solution to any problem, it is your responsibility to make clear which theorem you are using and to justify its use.

Problem 1. Let X_n be a sequence of variables that converge to X in probability. Show that there exists a subsequence X_{n_i} that converges to X almost surely.

Problem 2. Let X_i be a collection of independent, identically distributed random variables with mean 0, $\text{Var}(X_i) = 1$ and $\mathbb{E}[(X_i^2 - 1)^2] = 3$. What is the limiting distribution of

$$\frac{\sum_{i=1}^n X_{2i-1}}{\sqrt{n}} + \frac{\sum_{i=1}^n (X_{2i}^2 - 1)}{\sqrt{n}}$$

Problem 3. Let $B_t = (B_t^1, B_t^2, B_t^3)$ be a three-dimensional Brownian motion starting at the origin. Let τ be the first time that this Brownian motion hits the plane $\frac{1}{\sqrt{3}}[x + y + z] = 1$. Compute the probability $\mathbb{P}(\tau < 1)$.

Problem 4. We consider the following Markov process. We have two boxes A and B and a total of n balls split between the two boxes.

Let X_t be the random variable equal to the number of balls in urn A at time t . At time 0, we have X_0 balls in box A , and every second thereafter, we choose one of the n balls at random and shift it to the opposite urn.

- (1) Determine the invariant distribution of X .
- (2) Does the law of X_t converge to the invariant distribution regardless of the initial distribution of X_0 ?

Problem 5. Consider a Markov Process X_N given by

$$X_{N+1} = \begin{cases} X_N + X_N^2(1 - X_N)^3 \text{sign}(X_N - 1/2)w. & \text{probability } 2/3 \\ X_N - 2X_N^2(1 - X_N)^3 \text{sign}(X_N - 1/2)w. & \text{probability } 1/3 \end{cases}$$

with $X_0 = \frac{2}{3}$.

- (1) Show that X_N is a martingale.
- (2) Show that the sequence $\{X_N\}_{N=1}^{\infty}$ converges almost surely to some random variable X_{∞} and determine the distribution of X_{∞} .

Problem 6. Let ξ_i be a sequence of independent random variables(not necessarily identically distributed) on $\mathbb{Z}$ such that there exists some $p > 0$ with $\min(\mathbb{P}(\xi_i = 1), \mathbb{P}(\xi_i = -1)) > p$ for all i .

Consider the variable $S_n := \sum_{i=1}^n \xi_i$ and define $\tau_N := \inf\{i : |S_i| \geq N\}$.

- (1) Show that $\mathbb{E}[\tau_N] < \infty$.

Hint: Try to dominate τ_N by a geometric random variable.

- (2) Does the same conclusion hold if we do not assume ξ_i to be independent of each other?